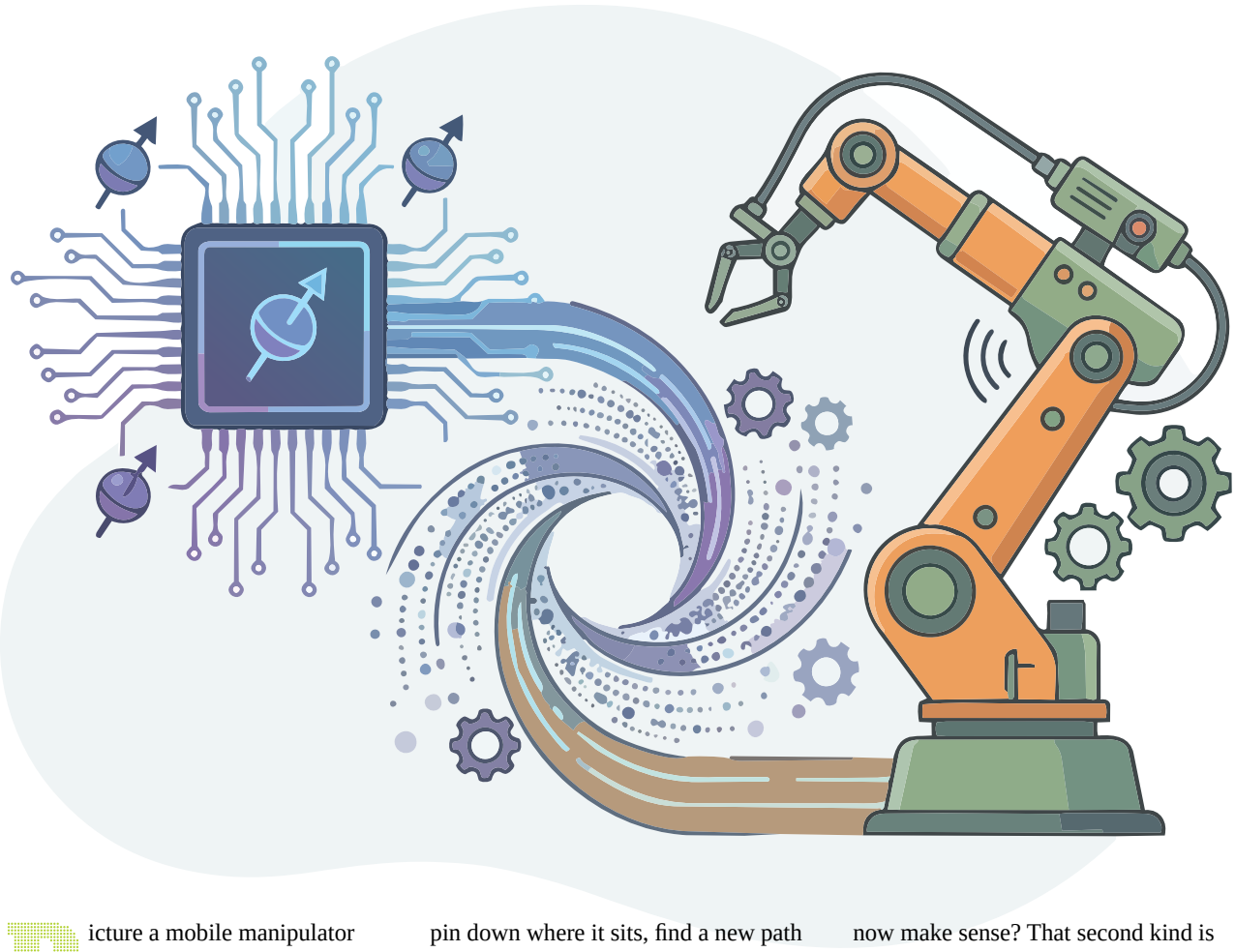
The Convergence of Quantum Computing and Physical AI

Industry can benefit immensely with quantum computing and Physical AI working together. That's what the tech is working towards.



Picture a mobile manipulator working the night shift in a parts warehouse. The scenario is imagined, but every piece of it is ordinary. The robot must pull forty items from scattered shelves, hand a few to a packing cell, and reach its charger before the battery dips below a safe margin. Halfway through, a pallet appears in an aisle that was clear five minutes earlier. The robot must notice it, pin down where it sits, find a new path without clipping a shelf upright, and keep its arm from swinging a heavy tote into anything as it turns.

Two different problems hide in that moment. One is fast and physical: keep the wheels and joints doing the right thing over the next few milliseconds. The other is combinatorial: with the aisle blocked, which order of picks, which route and which charging window now make sense? That second kind is where people start wondering whether a quantum computer could help. It might, for some slices of the problem, under conditions worth spelling out carefully.

Two fields that grew up apart

Physical AI is a loose label for machine learning systems that perceive and act in the physical world: robot arms, legged machines, drones and autonomous